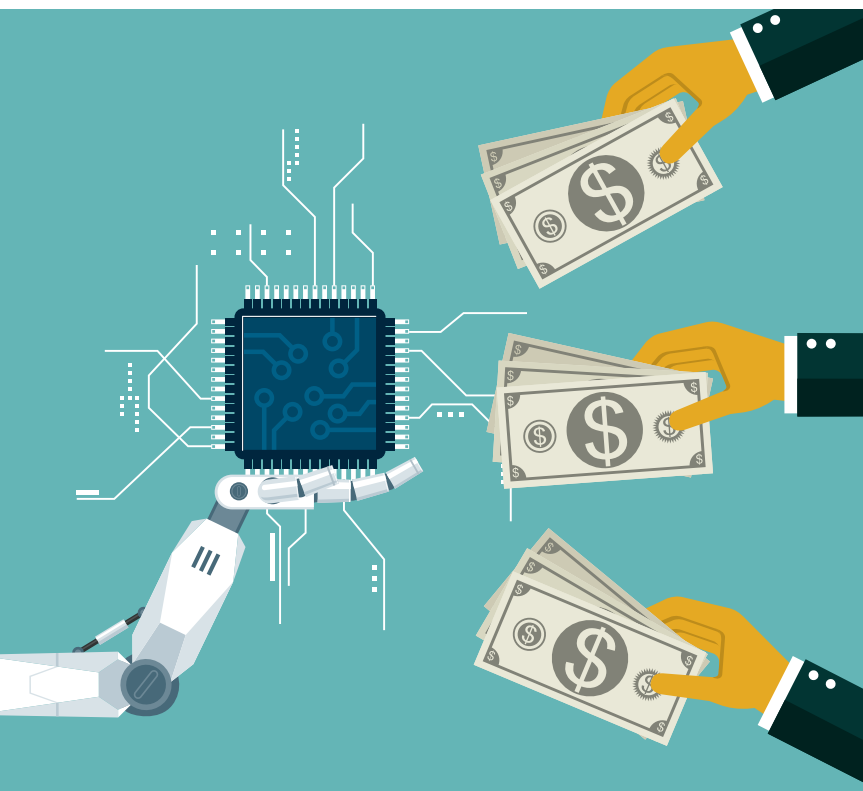




ATMPs: The Founding Stone Of India's Semiconductor Era

India has missed the semiconductor bus more than once in the past but with the likes of the United States, South Korea, and Taiwan announcing investments worth more than \$200 billion, the opportunity has once again come knocking on the doors. The semiconductors that these countries fabricate will need to be tested and packaged. Is India ready to grab this opportunity? India is best suited as a country for ATMPs and OSATs because of its design capabilities



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Maruti Suzuki, India's biggest car brand, has had to shut its operations for a while in the country. Similarly, the likes of Volkswagen, Mahindra, BMW, Mercedes Benz, and most other car makers had to either halt or feared possible disruption in their day-to-day

operations. The smartphone vertical, one of the hottest investment sectors in India, has also expressed their fear about the same. The common factor behind the fear was not Corona this time but the shortage of semiconductors!

As Prime Minister Narendra Modi has expressed time and again, if India is to succeed it will need to be *atmanirbhar* (self-reliant). This self-reliance factor weighs a little heavier for the electronics sector because India has very little on plate in terms of semiconductor manufacturing. Efforts have been made in the past, but nothing has taken shape yet. With the current government announcing production linked incentive (PLI) for electronics manufacturing ecosystem, hopes have started rising again. India's first bet in terms of semiconductor manufacturing might originate in the form of assembly, testing, marking, and packaging (ATMP) and outsourced semiconductor assembly and test (OSAT) units.

ATMP and OSAT units require lower investment in comparison to establishing semiconductor fabs, and at the same time can create job opportunities. Experts affirm that such units complement fabless semiconductor companies!

As Poornima Shenoy, former president of IESA and presently CEO of Hummingbird Advisors puts it, "Setting up an ATMP or OSAT could be the right step. The global market for ATMPs is estimated at \$32 billion. It is expected to grow to be worth \$180 billion by 2028. There are global ATMP firms registering nearly \$12 billion in revenue per year already!"